



The long road: Quantifying Europe's exposure to transport disruption

May 8, 2026

Heightened geopolitical risk in the Middle East has placed European transport networks under intense scrutiny, with a particular focus on the supply of jet fuel given the high share of the Gulf in the global supply of kerosene.

For the supply side of the European economy and overall input availability, the risk from a shortage of jet fuel is relatively small. The dominant fuel dependency for goods production is on diesel and electricity, reflecting the four-fifths share of input costs taken by land transport (road and rail).

The true vulnerability to a shortage of jet fuel lies squarely with the tourism sector. Our simulation shows a 10% drop in tourism flows would have a moderate impact on the Euro Area, on average (-0.2% of GDP), but a more significant impact on countries like Croatia (-1.5%), Greece, and Portugal (-0.7%).

Beyond the specific risk to jet fuel, a wider disruption to the global supply of transport would expose a more fundamental European vulnerability: the physical length of supply chains. To measure this vulnerability, we introduce a “physical distance travelled” metric, reflecting how far inputs travel for every industry. It reveals two critical points:

- A clear geographic divide: The economies with the longest supply chains are generally located in Southern Europe. Lacking deep integration with Europe's industrial core their value chains are inherently more global and reliant on long-distance sourcing. This exposure to transport frictions could create a vulnerability for some of the zone's high-debt member states. These countries are also those we previously found to have the highest energy import bill sensitivities.
- A dual "hidden" risk: From a sector perspective, the most-exposed sectors besides tourism are capital goods (electronics, air/spacecraft, machinery) and materials manufacturing (metals, refineries), whose supply chains stretch nearly 5,000 km, on average. Crucially, these are the same value chains we [previously identified](#) as having high "hidden" energy content, indicating a dual vulnerability to both transport shocks and embedded energy costs.

What does this mean for the ECB? Our baseline view is a 'measured' tightening with 50bp of hikes to 2.50%, the upper end of the range of neutral. With the labour market cooling, financial conditions tightening and this threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.

Authors

Yacine Rouimi
Senior Economist

Mark Wall
Chief Economist



Concerns shifting from energy to transport. Aviation of particular concern. As we highlighted in a recent [note](#), Europe's economic exposure to the Gulf conflict is twofold, and is more complex than the singular energy shock of the Russia-Ukraine war. Risks of direct shortages in energy and key materials like petrochemicals or mineral fuels are now coupled with a more complex threat: second-order disruptions to global value chains.

Transport disruption is one of the most significant vectors of transmission for both these risks. As an input at every stage of production, the transport sector's resilience is once again at the forefront amid reports of looming [fuel shortages and logistical bottlenecks](#). In all this, jet fuel is of particular concern, as shown by recent statements by the Head of the International Energy Agency on April 16 that Europe has "[maybe six weeks of jet fuel left](#)", and our own [calculations](#) that the share of Gulf countries as a source for EU imports of kerosene reaches 56%.

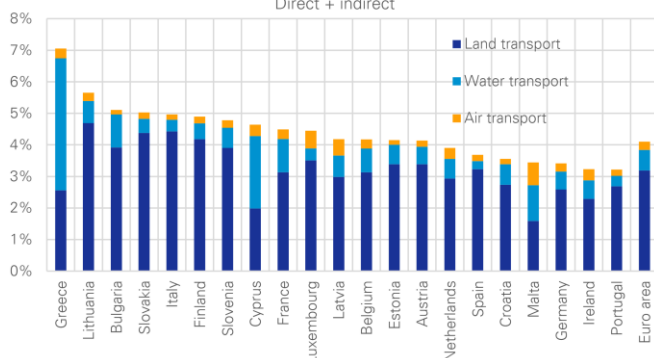
As our transportation sector specialists explain, the market is already under pressure. Jet fuel prices have doubled, and while replacement barrels from the US and Africa have helped Europe plug the gap for now, this fragile balance is unlikely to survive the summer demand surge. Airlines are already cancelling unprofitable flights and face difficult trade-offs between raising prices and eroding consumer confidence.

In this article, we think about the European economy's exposure to potential global transport disruption – in particular jet fuel – from two complementary perspectives. First, from a supply or production perspective, we look at the role of various forms of transport in European supply chains, including air transport, while introducing a distance metric reflecting how far European inputs travel and are exposed to transport disruptions. Second, from a demand or income perspective, we look at the exposure of European economies to tourism and in turn tourism's exposure to air travel.

How much transport in value chains? 3-5% of inputs, of which little air transport.
A traditional way to assess transport dependency is to measure its overall share in inputs along the value chain.

Figure 1: Cost share of transport services in Euro area input purchases, dominated by land. Air is tiny

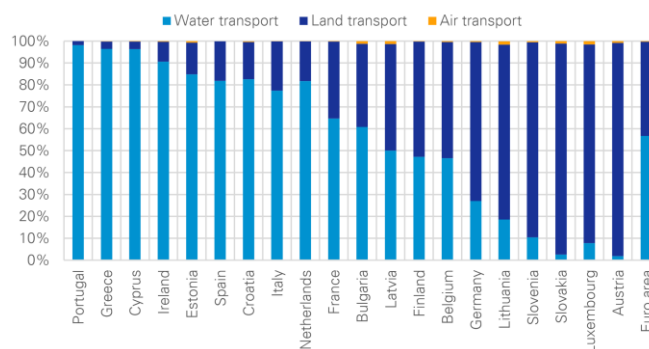
Euro area countries - Share of transport services in overall input costs - Direct + indirect



Source: Deutsche Bank Research, OECD ICIO. The figures include the absorption of all transport services at every stage of production, including that required to move inputs between two foreign suppliers (ie indirect absorption).

Figure 2: By physical weight, maritime shipping dominates freight transport

Euro area - Freight transport volume - By mode of transport



Source: Deutsche Bank Research, Eurostat (tran_hv_ms_frmod). The unit used for measuring the freight transport shares is the tonne-kilometre (tkm). One tonne-kilometre represents the movement of one tonne over a distance of one kilometre.



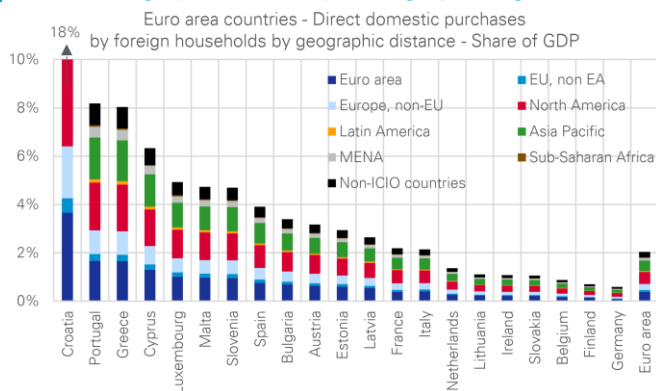
Our calculations, using a methodology [popularized by the OECD](#), suggest transport services account for 3-5% of total input purchases across the Euro area (Figure 1), with some notable outliers like Greece at 7%. Among the Big4, the role of transport is more pronounced in Italy and France (4.5-5%), less so in Germany and Spain (3-4%).

Land transport constitutes the bulk of this transport cost – approximately four-fifths. The share of water transport is significantly smaller, except for trade-reliant coastal nations like Greece, Malta, and Cyprus. Importantly, air transport represents only a minor share overall.

Implications for fuel usage. The dominance of land transport (i.e., road and rail) in the cost structure implies that the most significant fuel availability risk in the supply of goods and services comes from disruptions to diesel and electricity, not jet fuel. Given air freight's minimal share in both costs and physical volume of freight (as confirmed by both our analysis and Eurostat data in Figure 2), even a severe, un-substitutable 10% shortage of jet fuel may not cause much disruption to the flow of inputs and economic production in Europe.

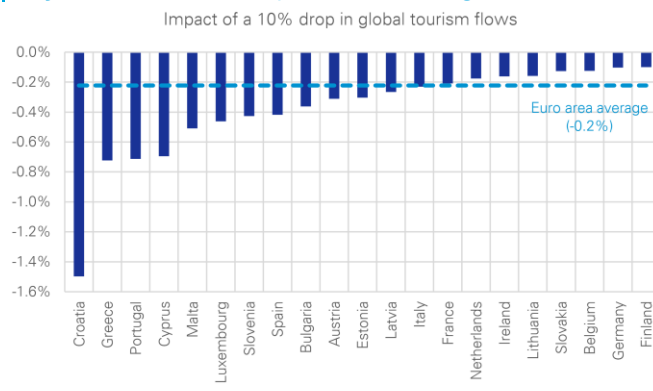
Tourism: a distinct, service-based vulnerability. The above numbers, focusing on freight and input purchases, do not capture the broader issue of the tourism economy: tourism involves moving individuals – not freight – and air transport is not counted as a formal input to the hotels, restaurants, and cultural activities that constitute the tourism economy (i.e., hospitality, recreation), but generally as a purchase by households.

Figure 3: The high share of long-distance travellers in direct foreign purchases (spending by foreign tourists)



Source: Deutsche Bank Research; OECD ICIO.

Figure 4: Tourism is where the Euro area's vulnerability to jet fuel and air transport becomes significant



Source: Deutsche Bank Research, OECD ICIO. The simulation focusses on the impact of spending by foreign tourists, and does not capture domestic tourism. As such, the numbers differ from the estimates of the sector's footprint published by Eurostat

In Figure 3, we calculate the share of GDP driven by direct purchases by foreigners (spending by foreign tourists). The result, unsurprisingly, is that the nations of Southern Europe are much more exposed, on average, but all nations in Europe derive income from a demand perspective from far-away visitors. We estimate that about three quarters of that spending is made by foreigners having travelled more than 2,000 km (50% of the spending by travellers more than 5,000 km away, often placing them on another continent).



A simulated 10% drop in global tourism flows reveals a manageable cost for the Euro Area, on average (a 0.2% hit to GDP – Figure 4). The impact is more pronounced in Croatia (GDP falling 1.5%), and Greece and Portugal (both seeing declines of 0.7%). On the Big4 countries, Spain would be faced with the most significant loss (0.4%), while France, Italy (0.2%) and Germany's (0.1%) exposures are more modest.

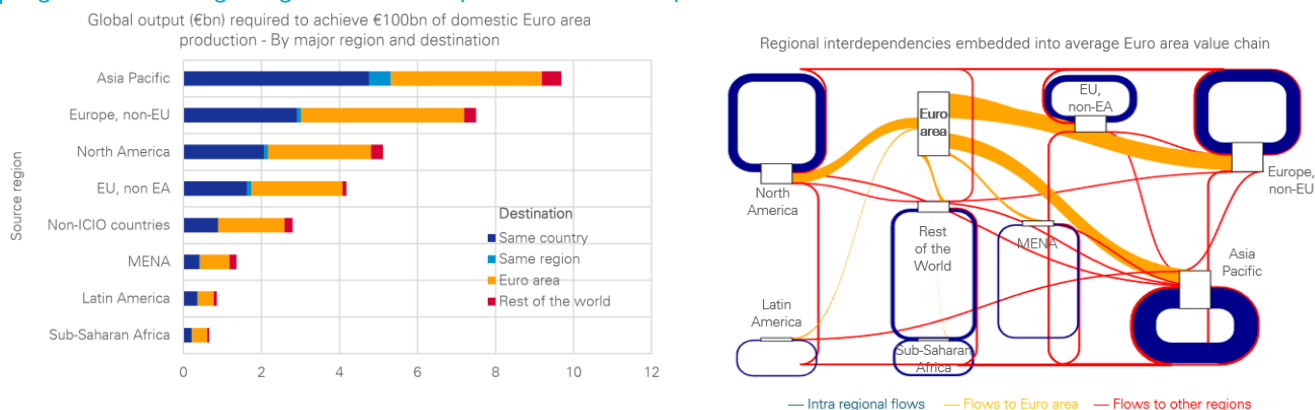
An important caveat to bear in mind: in cases of severe disruption of air travel, holidaymakers would likely travel more locally, which should reduce the size of a shock to the tourism industry. The overall impact will still depend on the size of the tourism sector relative to local demand.

Beyond jet fuel: Measuring exposures to transport friction. The focus on jet fuel, while critical for tourism, highlights a greater strategic risk: energy-driven transport disruptions may not stop at kerosene. A broader shock to transport fuels affecting diesel and maritime fuels would more certainly curtail the physical movement of goods. In such a scenario, what matters is not just the cost of transport, but a value chain's fundamental exposure to distance.

To assess this exposure, we introduce the “physical distance travelled” metric. This measure goes beyond a simple 'average import distance' to quantify the total physical journey embedded in the Euro Area's inputs to production, including domestic trade and all transactions that occur before a final good is imported. It reveals the true length – and vulnerability – of our supply chains. Deutsche Bank clients can access further details including the methodology on our [Deutsche Bank Research](#) client site.

Figure 5 illustrates the main input to the calculations. It shows the global transactions required to generate €100bn of production of an average good/service in the Euro Area, and where this output is directed. Both charts contain the same data, with the left-hand one showing how output is broken down (domestic consumption vs. exports), while the right-hand one illustrates the complex connections across regions to feed the Euro Area with inputs. The blue and red links represent the Euro Area's indirect exposure.

Figure 5: Picturing the global web of inputs for Euro area production



Source: Deutsche Bank Research, OECD ICIO. Note: our distance metric also includes the intra Euro area trade required in the value chain, not displayed in this chart.



We find a significant reliance on external partners, requiring approximately €10bn in output from the Asia-Pacific region alone. The charts also highlight the complexity of these value chains: a large portion of the Asian output is transacted and transformed internally (the blue segments) before being exported.

Tourism is clearly at risk, while the capital goods and materials value chains stand out on the goods-producing side. Southern Europe is more vulnerable, too.

The distance metric we calculate is simply the sum of the distances between the nodes in Figure 5 in kilometres, weighted by their values (it also includes intra euro area trade, not displayed in the chart). It shows that the longest, and, therefore, most transport-exposed, value chains in the Euro Area fall into three clear categories:

- First are tourism services, with most countries deriving a significant portion of revenue from travellers residing more than 5,000 km away (Figure 6).
- Second, Capital goods, where production is globally fragmented. Here, the computer and electronics sector stands out, with the average input traveling nearly 5,000 km, the longest of any European industry, after tourism. Other capital goods sectors like aircraft/railway equipment and electrical equipment (4,100 km) also exhibit noticeably long supply chains.
- The third category is materials manufacturing. Industries like non-ferrous metals (4,800 km), coke/refined petroleum (4,700 km), and iron/steel (4,000 km) all rely on sourcing raw materials from distant global locations. Notably, these are the same value chains we previously identified as having a high ["hidden energy"](#) imported content, indicating a dual vulnerability to both energy shocks and transport disruption.

Figure 6: Highest distance dependency for European tourism services, capital goods, and materials.

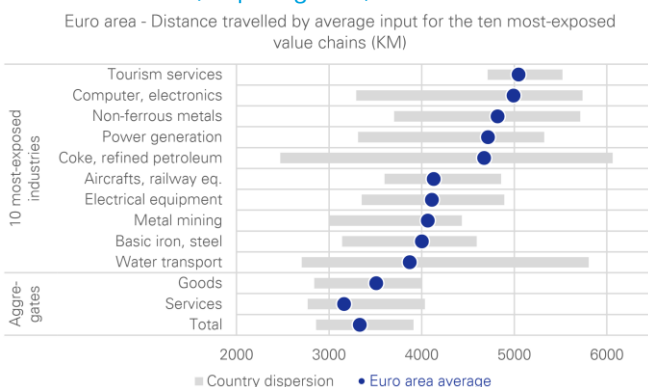
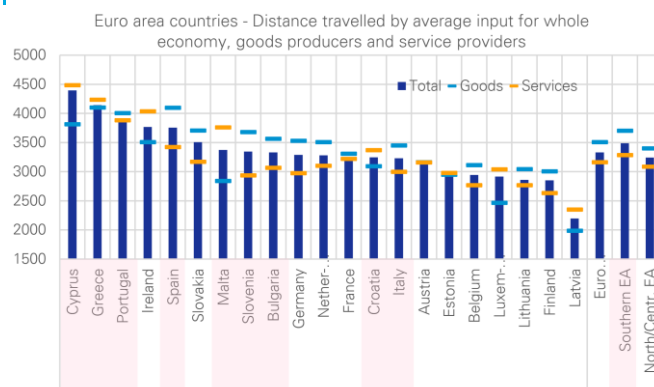


Figure 7: Longer distances in Southern European value chains



In terms of geographies, the simulations show that several Southern European countries have longer-than-average supply chains: a group including Cyprus, Greece, Portugal, and Spain are all positioned at the higher end of the spectrum (Figure 7). An average calculated for the nine Southern Euro area economies

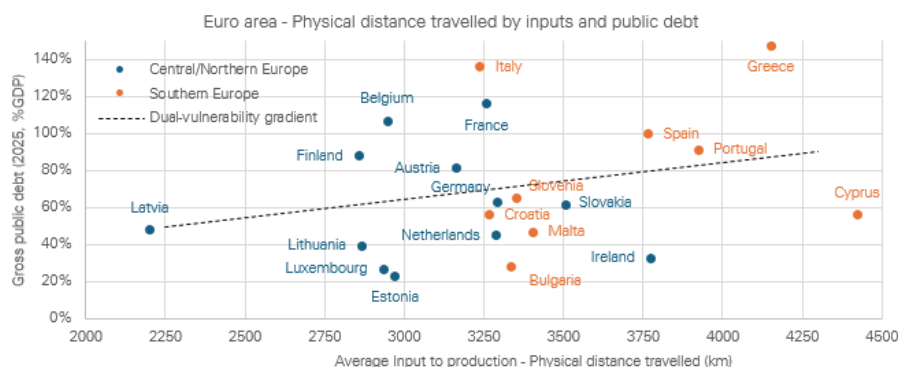


yields a kilometric distance of 3,490 km, a figure higher than the Euro Area average of 3,330 km. For goods producers within this southern bloc, the average distance is even longer, at 3,700 km.

The higher exposure to transport frictions could, therefore, create a vulnerability for some of the zone's high-debt member states (Figure 8). These countries are also those we previously found to have the highest energy import bill sensitivities.

Italy and Croatia are exceptions to this pattern, showing shorter supply chain distances more aligned with the industrial core of the Euro area, like Germany and France (3,200-3,300 km).

Figure 8: A vulnerability to transport disruptions for some of the highest-debt member States



Source: Deutsche Bank Research, OECD ICIO

What this means for the ECB: Measured tightening...at most

A negative supply shock pushes prices up and activity down. These opposing forces can neutralise the impact on monetary policy ('look through'). The ECB is nervous that the size and persistence of the upside inflation shock might need a tightening of monetary policy to ensure that inflation remains anchored at 2% in the medium term. However, the downside risks to growth are not irrelevant. The more significant the downside risks to growth, the less likely is a large or persistent inflation shock. Monetary policy needs to carefully weigh the inflation effects against the growth effects.

If Europe suffers from an outright shortage of transport fuels, then growth will undoubtedly weaken more than expected (assuming no substitutability of fuels or other sources of transport). Shortages could also push transport prices up even more, but only if firms believe demand can sustain even sharper price hikes, and this cannot be taken for granted – especially when the fiscal response to the crisis is limited and financial conditions are tightening.

In other words, if fuel supply shocks intensify to the point of outright shortages in Europe, it is not obvious that monetary policy would have to tighten even more than expected. It is even possible that monetary policy tightens less.



Our view is that the downside risks to growth are becoming more significant. One reason is the impact of the energy shock on supply chains. As discussed above, there are multiple aspects to this shock on supply chains.

First, we recently wrote about the hidden energy costs in Europe's imports, particular from Asia, a region where the risk of production disruptions from energy shortages is more acute.

Second, last year, we wrote about the high complexity (and low substitutability) of Europe's imports from Asia (China in particular). This increases the risk of an Asian shock compromising European supply chains.

Third, in this note, we offer a complementary perspective by showing the actual length ('physical distance travelled') of Europe's supply chains. The member states most at risk of an economic cost from a limited supply of transport fuels are those with longer supply chains. Generally speaking, they are also the ones with higher levels of public debt.

The bottom line is we see the net inflation/growth arguments favouring a measured ECB tightening at most. Our baseline is 50bp of hikes to 2.50%, with quarter point hikes in June and September. With the labour market cooling, financial conditions tightening and a threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.



Appendix 1

This material has been prepared by the Deutsche Bank Research Institute and is provided to you for general information purposes only. The Institute leverages the views, opinions, and research from Deutsche Bank Research, economists, strategists, and research analysts. Accordingly, you should assume that content in this document is based on or was previously published and provided to Deutsche Bank clients who may have already traded on the basis of it.

Any views or estimates expressed in this material reflect the current views of the author(s) and may differ from the views and estimates of Deutsche Bank AG, its affiliates, other Deutsche Bank personnel, and other materials published by Deutsche Bank. The content in this material is valid as of the date shown on the first page and may change without notice. Deutsche Bank has no obligation to provide any updates or changes to the information herein.

This material should not be used as a basis for trading securities or other financial products and should not be considered to be a recommendation or individual investment advice for any particular person. It does not constitute an offer, solicitation, or an invitation by or on behalf of Deutsche Bank to any person to buy or sell any security or financial instrument. Nothing in this material constitutes investment, legal, accounting or tax advice. Deutsche Bank engages in securities transactions, including on a proprietary basis, and may do so in a manner inconsistent with the views or information expressed in this material.

While information in this material has been obtained from sources believed to be reliable, neither Deutsche Bank AG nor any of its affiliates makes any representation or warranty, express or implied, as to the accuracy or completeness of the statements or any information contained in this material and therefore any liability is expressly disclaimed. This material is provided without any obligation, whether contractual or otherwise. Information regarding past transactions or performance is not indicative of future results.

In the US this report is approved and/or distributed by Deutsche Bank Securities Inc., a member of FINRA. In Germany this information is approved and/or communicated by Deutsche Bank AG Frankfurt, licensed to carry on banking business and to provide financial services under the supervision of the European Central Bank (ECB) and the German Federal Financial Supervisory Authority (BaFin). In the United Kingdom this information is approved and/or communicated by Deutsche Bank AG, London Branch, a member of the London Stock Exchange, authorized by UK's Prudential Regulation Authority (PRA) and subject to limited regulation by the UK's Financial Conduct Authority (FCA) (under number 150018) and by the PRA. This information is distributed in Hong Kong by Deutsche Bank AG, Hong Kong Branch, in Korea by Deutsche Securities Korea Co. and in Singapore by Deutsche Bank AG, Singapore Branch. In Japan this information is approved and/or distributed by Deutsche Securities Limited, Tokyo Branch. In Australia, retail clients should obtain a copy of a Product Disclosure Statement (PDS) relating to any financial product referred to in this report and consider the PDS before making any decision about whether to acquire the product.

By accessing this material, you agree that its content may not be reproduced, distributed or published by any person for any purpose, in whole or part, without Deutsche Bank's prior written consent. You also agree that you shall not scrape, extract, download, upload, sell or offer for sale any of the content in this material, and you agree not to use, or cause to be used, any computerized or other manual or automated program or mechanism, tool, or process, including any scraper or spider robot, to access, extract, download, scrape, data mine, display, transmit, or publish, any of the content in this material.